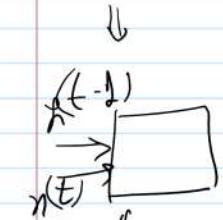
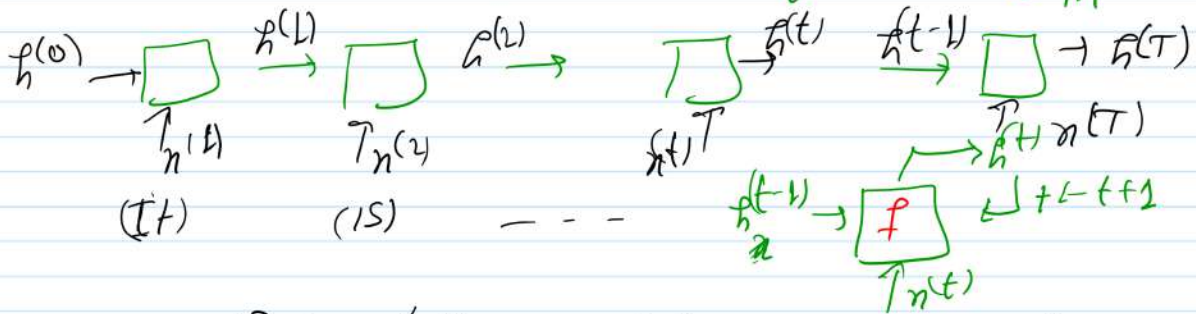


COL774 / WL734
Oct 28, 2025

It is good that COVID-19 has subsided
 T1 :- sentence (Pass, Li, Ah) $y \in \{1..C\}$
 T2 :- seq. labeling $y^{(1)} \dots y^{(T)}$
 T3 :- seq. to seq. mapping $\forall t: y^{(t)} \in \{1..C\}$
 $y^{(1)} \dots y^{(T)}$ $y^{(t')} \in \{1..C\}$



$$\Theta = (\Theta_F, \Theta_I)$$

$\hookrightarrow \mathbb{R}^{d \times d} \hookrightarrow \mathbb{R}^{d \times n}$

$n(t) \in \mathbb{R}^n$
 $h(t) \in \mathbb{R}^d \quad \forall t$

$$\begin{bmatrix} \Theta_F \\ \Theta_I \end{bmatrix} \begin{bmatrix} h^{(t-1)} \\ x^{(t)} \end{bmatrix} = h^{(t)}$$

$\hookrightarrow$ non-linearity applied pointwise

RNN:- Recurrent Neural Networks

$$y = \text{softmax}(\underbrace{\Theta_0}_{\text{parameters in output layer}} h^{(T)})$$

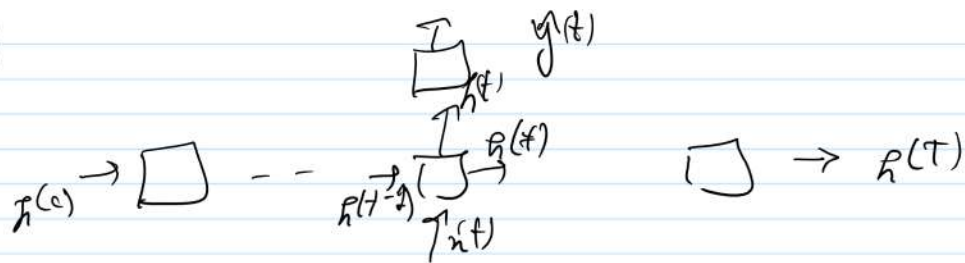
$$y = \text{softmax}(\underbrace{\Theta_0}_{\substack{\text{parameters in output layer} \\ \text{output}}} h^{(T)})$$

$$\Rightarrow \underbrace{\text{softmax}(a_1 \dots a_C)}_{\substack{\text{output} \\ y_1 \dots y_C}} = \frac{e^{a_c}}{\sum_{c'=1}^C e^{a_{c'}}}$$

$$\frac{1}{m} \sum_{i=1}^m \text{loss}(y_i^{(1)}, y_i^{(c)}) = \frac{1}{m} \sum_{i=1}^m \sum_{c=1}^C -1\{y_i^{(1)} = c\} \cdot \log y_{ic}^{(1)}$$

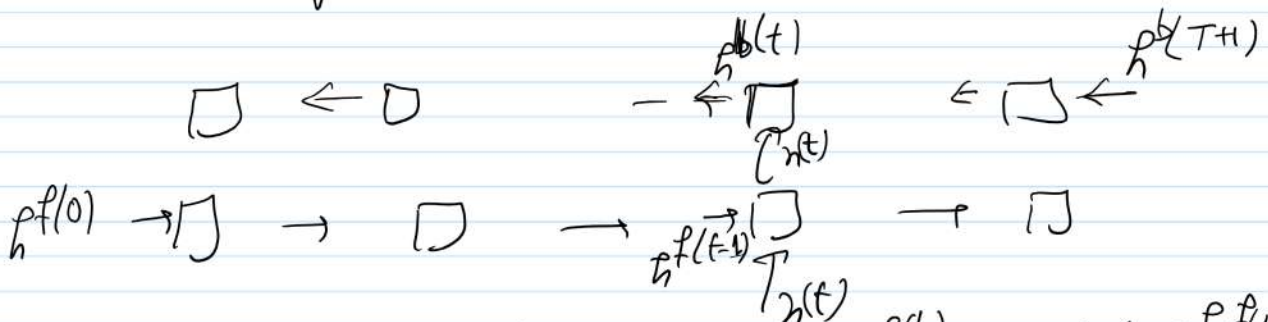
$$\text{Cross-Entropy}(P, Q) = - \sum_y P(y=y) \log Q(y=y)$$

(4)



$$h^{(t)} = g(\theta_h h^{(t-1)} + \theta_x x^{(t)})$$

$$y^{(t)} = \text{softmax}(\theta_o h^{(t)}) \quad \forall t$$



$$h^{(t)} = \begin{bmatrix} h^f(t) \\ h^b(t) \end{bmatrix}$$

$$y^{(t)} = g(\theta_o h^{(t)})$$

$$h^f(t) = g(\theta_h^f h^{f(t-1)} + \theta_x^f x^{(t)})$$

$$h^b(t) = g(\theta_h^b h^{b(t+1)} + \theta_x^b x^{(t)})$$

$$\text{Loss}(y^u, y^v) = \sum_i \sum_{t=0}^T \text{Loss}(y^{u(t)}, y^{v(t)})$$